

Package #136

Implement multi-modal sensing system within battery pack using optical sensors and acoustic monitors to detect early signs of thermal runaway in individual cells

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'traction battery thermal runaway initiation' in electric vehicles: 5 of 25 NHTSA owner complaints retrieved live at benchmark time (2021–2021) involve fire or battery-related faults. Narratives: 'The high voltage battery on my 2021 Tesla Model 3 Standard Range suddenly failed without any prior warning or gradual degradation. The vehicle was consistently achieving approximately 180 miles of range per full charge. '; 'This is my fourth high voltage battery failure on this vehicle. Battery replacements occurred at approximately 13,000 miles, 60,000 miles, 96,000 miles, and now at 121,000 miles. The first failure occurred at 13,000 mile'. Energy R&D; context: 10 DOE OSTI records on thermal-runaway propagation. LIMITS STAMPED INLINE: voluntary unverified reports, causality UNVERIFIED, counts are NOT incidence (Art. XXI.5 generalized).

Mechanism: Advanced characterization techniques including X-ray imaging, optical sensing, acoustic field imaging, and spectroscopy to study thermal runaway (TR) mechanism

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.